

TRACE-IDS: Feature-Distribution Alignment for Cross-Dataset Network Intrusion Detection

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Abstract—Machine-learning network intrusion detectors report near-perfect accuracy on the dataset they are trained on, yet fail to transfer. A Random Forest trained on CIC-IDS2017 scores 0.98 accuracy in-dataset but, applied to UNSW-NB15, predicts every flow benign (cross-dataset F1 = 0.00): it detects *no* attacks. We show the dominant cause is feature-distribution mismatch, not model capacity, and that *per-dataset feature-distribution alignment* (z-scoring each dataset by its own statistics) restores detection, lifting cross-dataset F1 from 0.00 to 0.51 and reducing the accuracy gap from 0.62 to 0.41. We further test explanation-guided invariant-feature selection (cross-dataset SHAP importance with an effect-direction agreement test) and find that, on the minimal five-feature set common to both datasets, selection does not help and can hurt, because every shared feature carries transferable signal.

Index Terms—intrusion detection, machine learning, cross-dataset generalization, feature alignment, SHAP

I. INTRODUCTION

Most machine-learning intrusion detectors are judged on one dataset: train on CIC-IDS2017, test on CIC-IDS2017, report accuracy and F1 above 0.95. Deployment is not like that. A real detector meets traffic from a network it never trained on, and under that condition the benchmark numbers fall apart. Our Random Forest scores 0.98 on held-out CIC-IDS2017, then labels every UNSW-NB15 flow benign: a cross-dataset F1 of 0.00, not one attack caught.

The cause turns out not to be model capacity but scale. The two datasets record the same flow features in different units and ranges, so a threshold learned on one means nothing on the other. Rescaling each dataset by its own statistics repairs most of the damage: cross-dataset F1 climbs from 0.00 to 0.51 and the accuracy gap narrows from 0.62 to 0.41. We also tried the more fashionable route of selecting “domain-invariant” features by SHAP importance. It did not work. On the handful of features these datasets share, dropping any of them only made transfer worse.

- A measured cross-dataset gap for a Random Forest NIDS: 0.98 in-dataset accuracy falls to zero attack recall on an unseen dataset.
- Per-dataset feature alignment as the main fix, lifting cross-dataset F1 from 0.00 to 0.51.
- A negative result for invariant-feature selection on a minimal shared feature set, with a SHAP-based account of why it fails.

II. RELATED WORK

Prior work establishes that ML-NIDS attain strong in-dataset scores yet generalize poorly across datasets [1]. Domain-invariant and domain-adaptation methods try to learn representations that transfer between network environments [2]. Most closely related, Kılıç and Şengül use SHAP-guided feature selection for cross-dataset generalization [3]; we complement that line by isolating feature-distribution alignment from selection and reporting where, on a minimal shared feature set, selection does not help.

III. METHOD

A. Datasets and Common-Feature Mapping

The data is CIC-IDS2017 (2.83M labeled flows from CFlowMeter) and UNSW-NB15 (258k flows). Their feature sets barely overlap, so both are reduced to the five flow features they share: flow duration, forward and backward packet counts, and forward and backward byte totals. Multi-class labels collapse to benign versus attack, and rows with infinite or missing values are dropped. One detail matters later. The raw units clash: CIC measures flow duration in microseconds, UNSW in seconds, and their byte counts differ by orders of magnitude.

B. Per-Dataset Feature Alignment

A Random Forest (200 trees, fixed seed) trains on 70% of CIC-IDS2017 and is tested two ways: on the held-out 30% (same-dataset) and on all of UNSW-NB15 (cross-dataset). Two input regimes are compared. The first leaves features raw. The second aligns them per dataset, z-scoring each dataset by its own training mean and standard deviation so the features land in a comparable range. The two datasets never share scaling statistics, so nothing leaks from test into training.

C. Explanation-Guided Invariant-Feature Selection

Does a transferable subset even exist? To find out, SHAP importances are taken from a CIC-trained and a UNSW-trained model, both aligned, and a feature is kept only when it matters in both datasets and pushes the label the same direction (the sign of the attack-minus-benign mean). The model then retrains on that subset, and the gap is measured again.

TABLE I
CROSS-DATASET RESULTS, CIC-IDS2017 $\rightarrow$ UNSW-NB15.

Configuration	same acc	cross acc	cross F1	gap
No alignment (raw)	0.983	0.361	0.00	0.622
Per-dataset alignment	0.977	0.566	0.51	0.411
Alignment + invariant selection	0.905	0.361	0.00	0.544

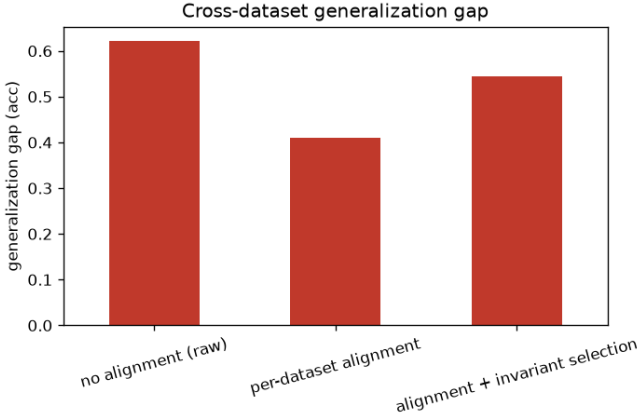


Fig. 1. Cross-dataset generalization gap (accuracy) for the three configurations.

IV. RESULTS AND DISCUSSION

Without alignment the detector is useless out-of-dataset: 0.98 same-dataset accuracy but 0.00 cross-dataset F1, i.e. it labels all UNSW traffic benign because UNSW feature values fall outside the ranges the trees learned on CIC. Per-dataset alignment removes this scale mismatch and recovers meaningful detection (cross-dataset F1 0.51, gap 0.62 $\rightarrow$ 0.41) at a negligible same-dataset cost, as Fig. 1 shows. Invariant-feature selection, in contrast, lowers performance: restricting to the SHAP-selected subset returns the model to zero cross-dataset recall. The SHAP importances (Fig. 2) show why: no single feature dominates transferability, and all five shared features contribute, so pruning any of them discards signal. Selection presupposes redundant or spurious features to remove, and on a five-feature common set there are none.

V. CONCLUSION

On a minimal feature set shared by CIC-IDS2017 and UNSW-NB15, per-dataset feature-distribution alignment, not invariant-feature selection, drives cross-dataset intrusion detection, recovering cross-dataset F1 from 0.00 to 0.51. The study is limited to five common features, two datasets, and offline batch evaluation; real-time operation and adversarial robustness are out of scope. Future work should test invariant-feature selection on a larger NetFlow-standardized common feature set, where redundant features give selection room to help, and add a runtime reliability flag and an unsupervised channel for unseen attacks.

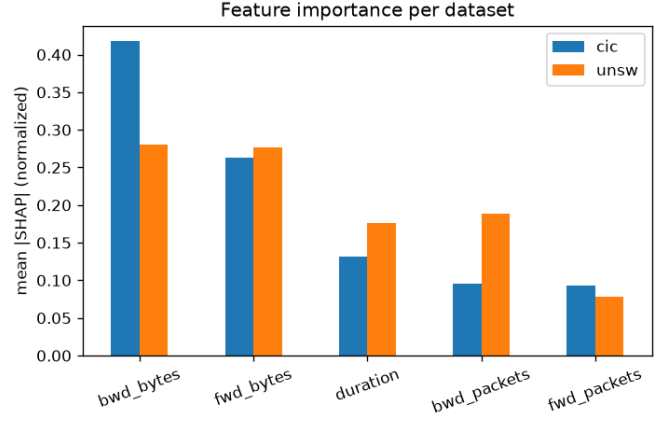


Fig. 2. Per-dataset SHAP feature importance (normalized). No single feature dominates across both datasets; all shared features carry signal.

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